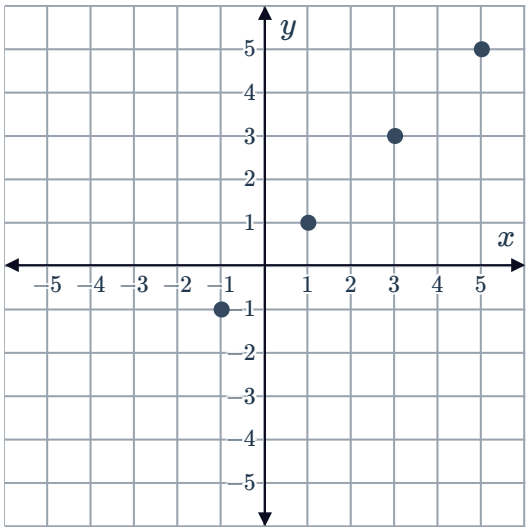
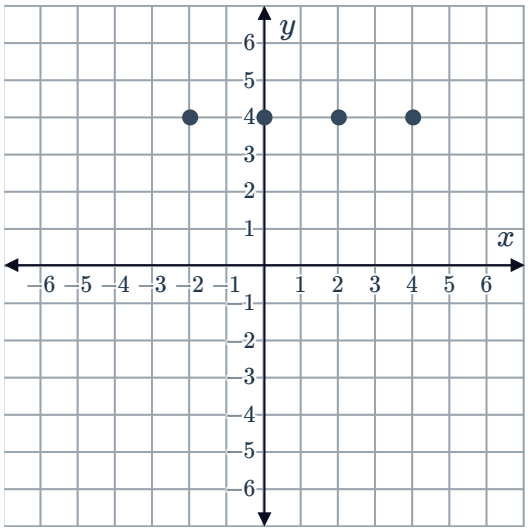
Linear relationships

1 Do the points on each of the following number planes form a linear relationship? Explain your answer.

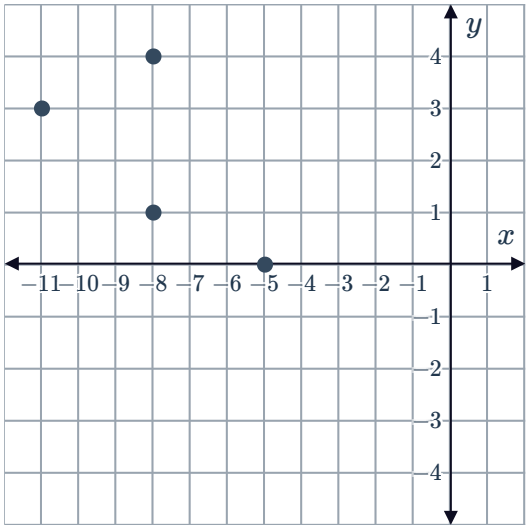
a



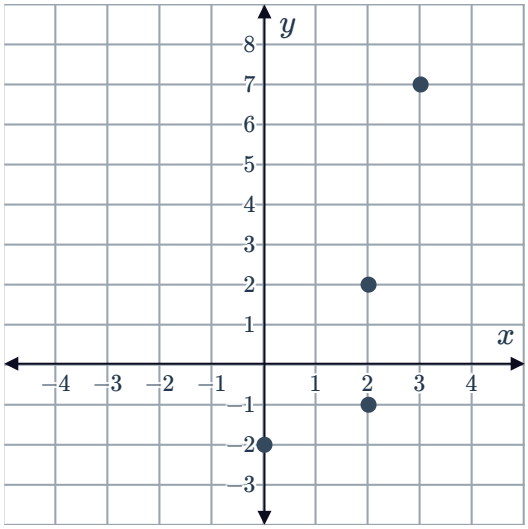
b



c



d



2 Consider the three points $A\,(3,6)$, $B\,(1,0)$, and $C\,(2,3)$.

a Plot the points on a number plane.

b Do the points on each of the following number planes form a linear relationship? Explain your answer.

3 Consider the three points $X\,(3,-7)$, $Y\,(4,-9)$, and $Z\,(1,-2)$.

a Plot the points on the same number plane.

b Do the points on each of the following number planes form a linear relationship? Explain your answer.

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Equations, tables and graphs

4 For each of the following equations and their table of values:

i Plot the points from the table on a number plane.

ii Determine whether the points form a linear relationship.

a $y = x + 5$

x	1	2	3	4
y	6	7	8	9

b $y = x$

x	1	2	3	4
y	1	2	3	4

c $y = 3$

x	-2	-1	0	1
y	3	3	3	3

d $y = x^2$

x	-2	-1	0	1
y	4	1	0	1

e $y = x + 4$

x	-4	-3	-2	-1
y	0	1	2	3

f $x = -6$

x	-6	-6	-6	-6
y	-6	-3	0	3

g $y = -x$

x	-3	-2	-1	0	1
y	3	2	1	0	-1

h $y = 4x$

x	-2	-1	0	1	2
y	-8	-4	0	4	8

5 For each of the following equations:

i Complete the table of values provided.

ii Plot the points from the table on a number plane.

a $y = 3x$

x	-1	0	1	2
y				

b $y = x - 5$

x	4	5	6	7
y				

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c $y = x + 3$

x	1	3	7	10
y				

d $y = 2x + 7$

x	-2	-1	0	1
y				

e $y = 3$

x	-6	-3	-2	2
y				

f $y = -2x + 1$

x	-2	-1	0	1
y				

6 Consider the given table of values:

x	-3	-2	-1	0
y	15	10	5	0

a Plot the points in the table on a number plane.

b Is the relationship between x and y linear?

c If we extend the table of values to include $x = 1$, find the corresponding value of y.

7 Consider the given table of values:

x	-8	-4	0	4
y	-2	-1	0	1

a Plot the points in the table on a number plane.

b Is the relationship between x and y linear?

c If we extend the table of values to include $x = 8$, find the corresponding value of y.

8 Consider the given table of values:

x	-1	0	1	2
y	-3	1	5	9

a Plot the points in the table on a number plane.

b Is the relationship between x and y linear?

c If we extend the table of values to include $x = 3$, find the corresponding value of y.

9 Consider the given table of values:

x	-2	-1	0	1
y	2	-1	-4	-7

Does the ordered pair $(-4, 8)$ satisfy the linear relationship between x and y?

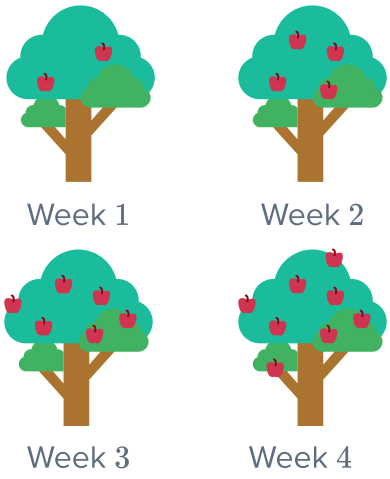
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Applications

10 After each week, the number of apples on an apple tree increases according to the diagram:

a Complete the following table of values:

Week	1	2	3	4
No. Apples				



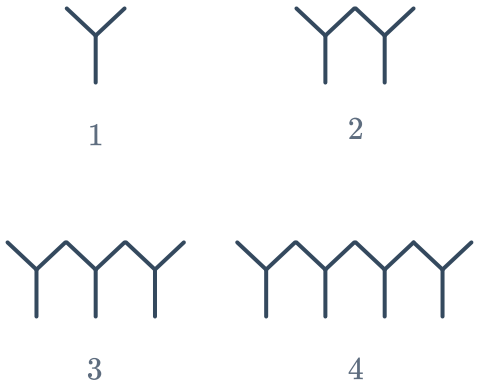
b Plot the number of apples against each week on a number plane.

c Is the relationship between the number of weeks and the number of apples linear?

11 The following pattern was created using matchsticks. Each figure is labelled with a number:

a Complete the following table of values:

Figure number	1	2	3	4
No. Matchsticks				

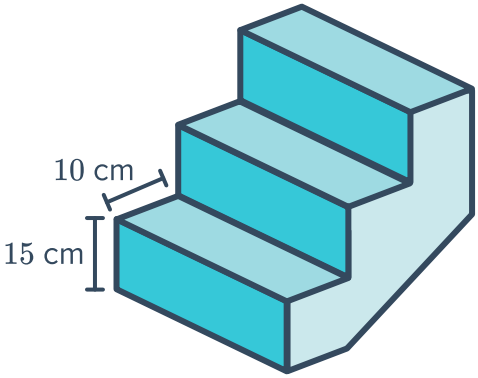


b Plot the number of matchsticks used at each step on a number plane.

c Is the relationship between the step number and the number of matchsticks linear?

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12 A staircase with 5 steps is shown below. The height of each step is 15 cm and the length of each step is 10 cm.



a Complete the following table of values:

No. Steps	2	3	4	5
Height of staircase (cm)	30			

b Plot the height of a staircase against the number of steps on a number plane.

c Is the relationship between the height of a staircase and the number of steps linear?

d Is it possible to have a staircase with height 105 cm with steps that have the same height and length as in the image?

e Complete the following table of values:

No. Steps	2	3	4	5
Length of staircase (cm)	20			

f Plot the length of a staircase against the number of steps on a number plane.

g Is the relationship between the length of a staircase and the number of steps in a staircase linear?

h Is it possible to have a staircase with length 70 cm with steps that have the same height and length as in the image?

13 A water tank initially starts with 65 L of water. Each day, 10 L of water is taken from the water tank.

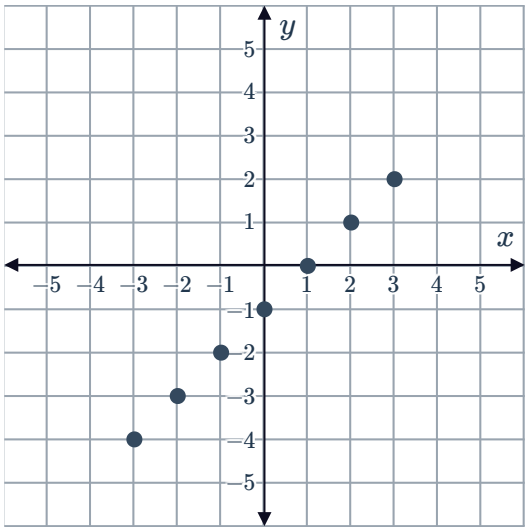
a Plot the volume (in L) of water in the tank against the number days, from 0 to 4 days.

b Is the relationship between the volume of water in the water tank and the number of daysd linear?

c How many litres of water will there be after 5 days?

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14 The points on the number plane show the relationship between water temperature, x, and surface air temperature, y:



a Complete the table of values:

Water Temperature in $^\circ$C, x	-3	-2	-1	0	1	2	3
Surface Air Temperature in $^\circ$C, y							

b Find the surface air temperature when the water temperature is 4°C.

c Find the surface air temperature when the water temperature is 14°C.

15 Let the height of a candle be h cm. If the candle is lit, the height of the candle decreases according to the equation $h = -2t + 9$, where t is the time in minutes. A table of values is shown:

Time (t mins)	0	1	2	3
Height of candle (h cm)	9	7	5	3

Plot the points in the table on a number plane.

16 A dead tree is 7 m tall and its height does not change over time. Let the height of the tree be y metres and the time passed be x years.

a State whether the following coordinates satisfy the growth of the tree, where the x-coordinate is the time passed, and the y-coordinate is the height of the tree:

i $(3, 6)$ **ii** $(0, 7)$ **iii** $(7, 21)$ **iv** $(10, 1)$

v $(4, 7)$ **vi** $(2, 7)$ **vii** $(7, 0)$ **viii** $(7, 7)$

b Plot the points from part (a) that satisfy the growth of the tree on a number plane.

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17 Let the volume of water in a tank be V L, and the time taken to fill up the tank be t minutes. Suppose the tank is initially empty and the number of litres in the tank is always three times the number of minutes that have passed.

a Complete the table of values:

Time (t mins)	0	1	2	3
Volume of water (V L)				

b Which of the following equations represents the relationship between the time taken, t, and the volume of the tank, V?

A $V = 3$ **B** $V = 3t$ **C** $V = 3t + 2$ **D** $V = t + 3$

c Plot the points in the table on a number plane.

18 Let the height of a pot plant be y cm. As the pot plant grows, the height increases according to the equation $y = 2t + 9$, where t is the elapsed time in weeks.

a Complete the table of values:

Time (t weeks)	0	1	2	3
Height of pot plant (y cm)				

b Plot the points in the table on a number plane.

19 Max is 21 years old and Sebastian is 14 years old. Let M represent Max's age and S represent Sebastian's age.

a Write an equation for M in terms of S.

b State whether following ordered pairs, in the form (S, M), satisfy the relationship between Max and Sebastian's ages:

i $(14, 21)$ **ii** $(36, 29)$ **iii** $(20, 27)$ **iv** $(25, 25)$

v $(24, 31)$ **vi** $(16, 23)$ **vii** $(38, 46)$

c Plot the points from part (b) that satisfy the relationship between their ages on a number plane.

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